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Research Interests

Research Area: Artificial Intelligence, Embodied AI, World Models, Vision-Language-Action (VLA) Model, Multi-Agent Systems

Education & Work Experience

	National University of Singapore Visiting PhD Student	2026–Present
	Sun Yat-Sen University – Phd Computer Science and Technology	2024–Present
	Huawei Technologies Co., Ltd. Hardware Engineer, Consumer Business Group	2023–2024
	Sun Yat-Sen University – Master Electronic Science and Technology	2020–2023
	China Agricultural University – Bachelor Electronic Information Engineering	2016–2020

Selected Publications & Preprints

Co- π -tree: Distilling LLM Reasoning into an Interpretable Policy Tree for Human-AI Collaboration B. Zhang, Y. Liang, Guowei Zou , H. Wang, H. Wu. EMNLP 2026	2026
CoFlow: Coordinated Few-Step Flow for Offline Multi-Agent Decision Making Guowei Zou , H. Wang, Beiwen Zhang, Boning Zhang, H. Wu. arXiv:2605.01457, 2026	2026
One Step Is Enough: Dispersive MeanFlow Policy Optimization (DMPO) Guowei Zou , H. Wang, H. Wu, Y. Qian, Y. Wang, W. Li. ACM Multimedia (ACM MM), 2026	2026
DM1: MeanFlow with Dispersive Regularization for 1-Step Robotic Manipulation Guowei Zou , H. Wang, H. Wu, Y. Qian, Y. Wang, W. Li. arXiv:2510.07865, 2025	2025
D ² PPO: Diffusion Policy Policy Optimization with Dispersive Loss Guowei Zou , W. Li, H. Wu, Y. Qian, Y. Wang, H. Wang. AAAI 2026	2025

Additional Publications

Modeling light propagation for under-display sensing in a smartphone Q. Wang, Y. Liu, Guowei Zou , Z. Liang, et al. Optics Express	2025
18-3: Mini-LED LCDs Integrated with High-capacity MIMO Visible Light Communication Z. Zhao, Y. Qiu, Guowei Zou , J. Weng, B.-R. Yang, Z. Qin. SID Symposium Digest of Technical Papers	2024
High-capacity MIMO visible light communication integrated into mini-LED LCDs Z. Zhao, Y. Qiu, Guowei Zou , Y. Liu, et al. Optics Express	2024
Enhancing the spatial resolution of light-field displays without losing angular resolution by a computational subpixel realignment W. Yang, Y. Cheng, Guowei Zou , B.-R. Yang, Z. Qin. Optics Letters	2023

Two-field sequential color liquid crystal displays with deep learning-enabled real-time driving Z. Wang, Guowei Zou , Y. Li, C. Ma, et al. Optics Letters	2023
Deep learning-based real-time driving for 3-field sequential color displays with low color breakup and high fidelity Z. Wang, Guowei Zou , Y. Shen, B.-R. Yang, Z. Qin. Optics Express	2023
Deep Learning Designed Driving Method for Field Sequential Color Displays Z. Wang, Guowei Zou , Y. Shen, B.-R. Yang, Z. Qin. Laser Science	2022
75-3: Field sequential color displays with simultaneously suppressed color breakup and flicker based on multi-objective optimization Z. Wang, Guowei Zou , B.-R. Yang, Z. Qin. SID Symposium Digest of Technical Papers	2022
65-1: Deep Learning-Enabled Image Content Adaptive Driving Algorithm for Field Sequential Color LCDs with Mini-LED Backlight Guowei Zou , Z. Wang, W. Yang, B.-R. Yang, Z. Qin. SID Symposium Digest of Technical Papers	2022
42-2: Spatial resolution-tripled integral imaging light field displays with no loss of angular resolution by recombining subpixels with zero sampling error W. Yang, Y. Cheng, Guowei Zou , B.-R. Yang, Z. Qin. SID Symposium Digest of Technical Papers	2022
Deep learning-enabled image content-adaptive field sequential color LCDs with mini-LED backlight Guowei Zou , Z. Wang, Y. Liu, J. Li, et al. Optics Express	2022

Granted Patents

Visible Light Communication Method, Device and System Using Field Sequential Color LCD. Z. Qin, Z. Zhao, Guowei Zou , Y. Qiu, B.-R. Yang.	Granted 2026 ZL202211458797.0
Adaptive Refresh Rate Field Sequential Color Display. Z. Qin, Guowei Zou , Z. Wang, H. Liang, B.-R. Yang.	Granted 2025 ZL202210380314.3
Deep Learning-based Control for Field Sequential Display. Z. Qin, Guowei Zou , Q. Luo, W. Yang, Z. Qiu, Z. Wu, B.-R. Yang.	Granted 2023 ZL202110793074.5
High-resolution Light Field Display Method. Z. Qin, W. Yang, Y. Cheng, Guowei Zou , Y. Gong, Z. Wu, B.-R. Yang.	Granted 2022 ZL202110873615.5

Internship Experience

 China Star Optoelectronics Technology Co., Ltd. (CSOT)	2022–2023
AI for image enhancement: computer-vision-based modeling and optimization for image quality improvement.	
 Huawei Technologies Co., Ltd.	2021–2023
VR/AR image enhancement: AI-based image restoration and visual quality improvement for AR-HUD systems.	
 Vivo Mobile Communication Co., Ltd.	2021–2023
Mobile vision sensing: deep learning-based sensor modeling and signal enhancement for mobile imaging.	
 Tridium Inc. (Honeywell)	2019–2020
AIoT smart building: sensor data analysis and data-driven automation for smart building systems.	
 Wlkata Robotics	2019–2020
Robotic arm control: motion planning and control algorithms for robotic manipulation.	

Honors & Awards

Distinguished Student Paper Award International Conference on Display Technology (ICDT), Hefei, China <i>Simulation methods for under-display sensing: Near-field and far-field light propagation.</i>	2024
Distinguished Student Paper Award International Conference on Display Technology (ICDT), Nanjing, China <i>Mini-LED LCDs Integrated with High-capacity MIMO Visible Light Communication.</i>	2023

Distinguished Student Paper Award International Conference on Display Technology (ICDT), Fuzhou, China <i>Simulation and analysis of edge ghosting for microcapsule e-paper based on particle dynamics and light scattering model.</i>	2022
Best Paper Award Cross-Strait Postgraduate Workshop on Display Technology (PGWS), Hong Kong, China	2021
Beijing Excellent Undergraduate Thesis Beijing Municipal Government Top 1%	2020
Excellent Undergraduate Thesis China Agricultural University Top 3%	2020

Selected Conference Presentations

The 34th ACM International Conference on Multimedia (ACM MM 2026) Rio de Janeiro, Brazil Upcoming	2026
AAAI Conference on Artificial Intelligence (AAAI-26) Singapore	2026
Society for Information Display (SID) Display Week California, USA Oral presentation	2022
International Conference on Display Technology (ICDT) Fuzhou, China Oral presentation	2022
International Display Manufacturing Conference (IDMC) Taipei, China Poster presentation	2022
Cross-Strait Postgraduate Workshop on Display Technology (PGWS) Hong Kong, China Oral presentation	2021